RIYADH FLY

FLIGHT RECORDS ANALYTICS

RUH · KING KHALID INTERNATIONAL AIRPORT

Riyadh Fly

Flight Data Analysis and Predictive Insights

PREPARED BY

Sulaiman Allumaylim

Table of Contents

01 Introduction 03

02 Problem Definition 04

03 Dataset Description 05

04 Research Questions and Expected Insights 06

05 Data Exploration 06

06 Data Cleaning 07

07 Exploratory Data Analysis (EDA) 08

08 Dashboard Overview 10

09 Key Insights 12

10 What I Learned 12

11 Limitations of the Dataset 13

12 Recommendations for Future Analysis 13

13 Conclusion 14

14 References 14

— SECTION 01

Introduction

King Khalid International Airport (RUH) is a major aviation hub in Riyadh, Saudi Arabia, handling domestic and international flight movements. Understanding flight activity is important for identifying operational patterns, evaluating airline activity, and examining how flight volumes vary over time.

The Riyadh Fly project analyzes flight records from the King Khalid International Airport Flights Dataset, obtained from Kaggle. The analysis focuses on the second and third quarters of 2025 (April–September), using data cleaning, exploratory data analysis (EDA), and data visualization techniques to examine monthly flight activity, the most frequently recorded destinations, airline activity, and peak flight hours.

To make the results accessible and easy to interpret, the project includes an interactive dashboard developed using Streamlit. The dashboard presents historical flight data through visualizations and organized sections, allowing users to explore key patterns and compare flight activity across different periods.

In addition to analyzing historical patterns, the project uses predictive modeling to estimate flight activity for the fourth quarter of 2025 (October–December). A Random Forest Regressor is used to forecast daily flight counts and hourly flight activity, while historical averages are used to estimate the most active destinations and airlines. These forecasts are also presented in the Streamlit dashboard, providing a unified view of historical analysis and predicted trends.

Overall, Riyadh Fly aims to transform raw flight records into meaningful insights through data analysis, predictive modeling, and interactive visualization, supporting a clearer understanding of flight activity at King Khalid International Airport.

— SECTION 02

Problem Definition

Airports generate large volumes of flight data, making it challenging to identify meaningful patterns and understand changes in flight activity without proper data analysis and visualization. Raw flight records alone do not provide a clear overview of monthly flight volumes, frequently served destinations, airline activity, or the hours with the highest flight activity.

For King Khalid International Airport, analyzing historical flight records can help reveal these patterns and provide a better understanding of flight activity over time. However, the data must first be cleaned, organized, and analyzed to ensure that the results are meaningful and suitable for interpretation.

Another challenge is estimating future flight activity based on historical patterns. Understanding potential changes in flight volumes, airline activity, destinations, and busy hours requires appropriate analytical and predictive techniques.

Therefore, the problem addressed by the Riyadh Fly project is how to transform historical flight data into meaningful insights and estimates of future activity through data cleaning, exploratory data analysis, predictive modeling, and an interactive Streamlit dashboard. The project focuses on analyzing flight records from April to September 2025 and estimating flight activity for October to December 2025, presenting the findings through clear visualizations that make the results easier to understand and explore.

— SECTION 03

Dataset Description

DATASET TITLE	King Khalid International Airport Flights Dataset
SOURCE	Kaggle – King Khalid International Airport Flights Dataset

SHORT DESCRIPTION

The dataset contains flight records associated with King Khalid International Airport (RUH) in Riyadh, Saudi Arabia. It includes information such as flight numbers, airline names, aircraft models, flight status, flight types, origin and destination airports, terminals, and scheduled flight times. The original dataset contains 153,308 records covering the period from March 15 to October 10, 2025. For this project, the analysis focuses on flights recorded between April and September 2025, corresponding to the second and third quarters of the year.

REASON FOR SELECTION

This dataset was selected because it provides relevant information for analyzing flight activity at a major Saudi Arabian airport. In addition to its analytical value, the aviation industry is an interesting and fascinating field, particularly because of its complexity, global connectivity, and the continuous movement of flights between destinations. This interest motivated the selection of a dataset that combines an engaging subject with practical opportunities to apply data analysis and machine learning techniques.

The dataset's temporal, airline, aircraft, and destination attributes make it suitable for exploratory data analysis, identifying flight activity patterns, comparing quarterly performance, and developing predictive models. Furthermore, it supports the development of an interactive Streamlit dashboard that presents historical insights and estimated flight activity for the fourth quarter of 2025.

— SECTION 04

Research Questions and Expected Insights

MAIN QUESTIONS TO EXPLORE

This project explores how flight activity varies across Q2 and Q3 2025, which destinations and airlines record the highest number of flights, and which hours are the busiest. It also examines what flight activity patterns can be expected in Q4 2025 based on historical data.

EXPECTED INSIGHTS

The project aims to identify monthly flight patterns, the most active destinations and airlines, and peak flight hours. It also estimates Q4 2025 flight activity using predictive modeling and historical averages. The findings are presented through an interactive Streamlit dashboard to make the results easier to understand and explore.

— SECTION 05

Data Exploration

The dataset was explored to understand its structure, identify data quality issues, and discover flight activity patterns. The analysis focused on monthly flight counts, the most frequently served destinations, the most active airlines, and the busiest hours of the day. Visualizations were used to compare flight activity across Q2 and Q3 2025 and identify key trends. These findings also provided a foundation for developing the Q4 2025 forecasts and presenting the results through the interactive Streamlit dashboard.

— SECTION 06

Data Cleaning

The dataset was cleaned and prepared through the following steps:

01 **First, selecting relevant columns:** Only the columns required for flight analysis and dashboard development were retained.

02 **Second, converting data types:** The scheduled movement time column was converted to the datetime format to enable accurate date and time processing.

03 **Third, filtering the date range:** Only flight records from Q2 and Q3 of 2025 (April–September) were retained to ensure a consistent analysis period.

04 **Fourth, removing duplicates:** Duplicate rows were identified and removed. The dataset size was compared before and after this step to verify the impact of duplicate removal.

05 **Fifth, splitting date and time:** The scheduled movement timestamp was separated into individual date and time columns, making it easier to analyze daily and hourly flight patterns.

06 **Finally, renaming columns:** Column names were standardized and simplified to improve readability and usability in Python, data analysis, and dashboard development.

ORIGINAL RECORDS	ORIGINAL COLUMNS	CLEANED RECORDS	CLEANED COLUMNS
153,308	23	135,456	7

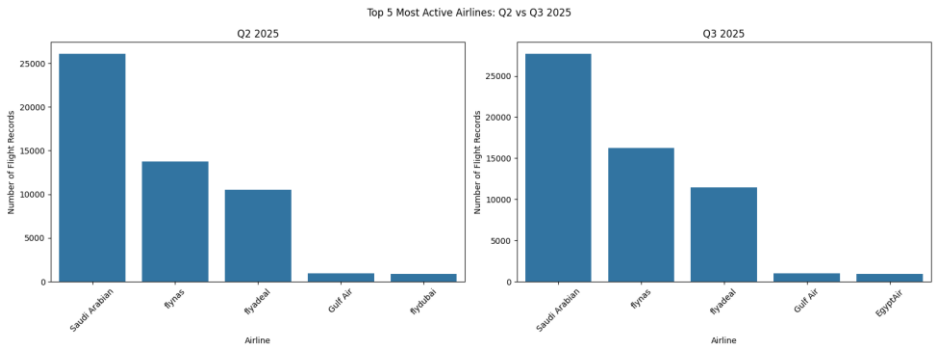
After completing the cleaning process, the dataset was reduced from 153,308 records and 23 columns to 135,456 records and 7 columns. The final dataset was saved as Riyadh_Fly_EDA_Cleaned.csv and used for exploratory data analysis, predictive modeling, and the interactive Streamlit dashboard.

— SECTION 07

Exploratory Data Analysis (EDA)

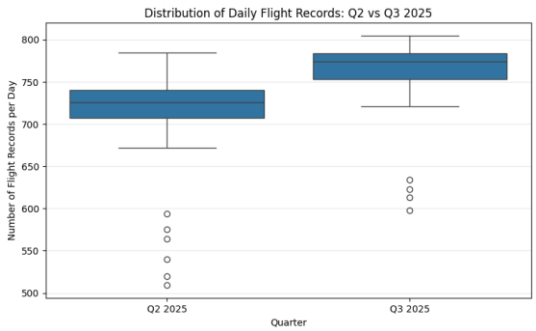
Exploratory Data Analysis (EDA) was conducted to identify patterns and understand flight activity at King Khalid International Airport during Q2 and Q3 of 2025. Several visualizations were created to examine monthly flight volumes, the most frequently served destinations, the most active airlines, and the busiest hours of the day. Comparing these patterns across the two quarters helped highlight changes in flight activity over time. The analysis showed that Jeddah was the most frequently served destination, Saudia was the most active airline, and flight activity generally increased during the afternoon and evening.

VISUALIZATION 1 **Top 5 Most Active Airlines**



The chart compares the five most active airlines in the analyzed dataset.

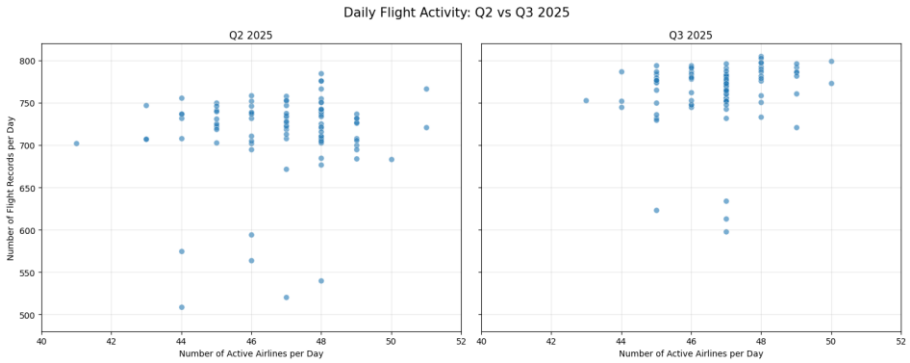
VISUALIZATION 2 **Boxplots**



Q3 had a higher median number of daily flight records than Q2. Daily flight counts were more concentrated in Q3, while both quarters had some unusually low-count days.

VISUALIZATION 3

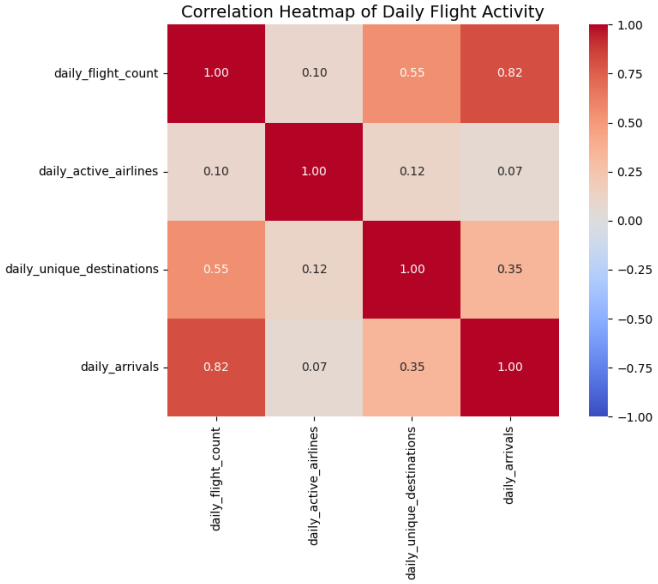
Scatter Plots



Most days recorded approximately 700–800 flight records. However, the scatter plot does not show a clear pattern indicating that having more active airlines necessarily leads to more flight records.

VISUALIZATION 4

Correlation Heatmap



Daily flight records showed a strong positive correlation with arrivals (0.82) and a moderate correlation with unique destinations (0.55). The relationship with active airlines was weak (0.10).

— SECTION 08

Dashboard Overview

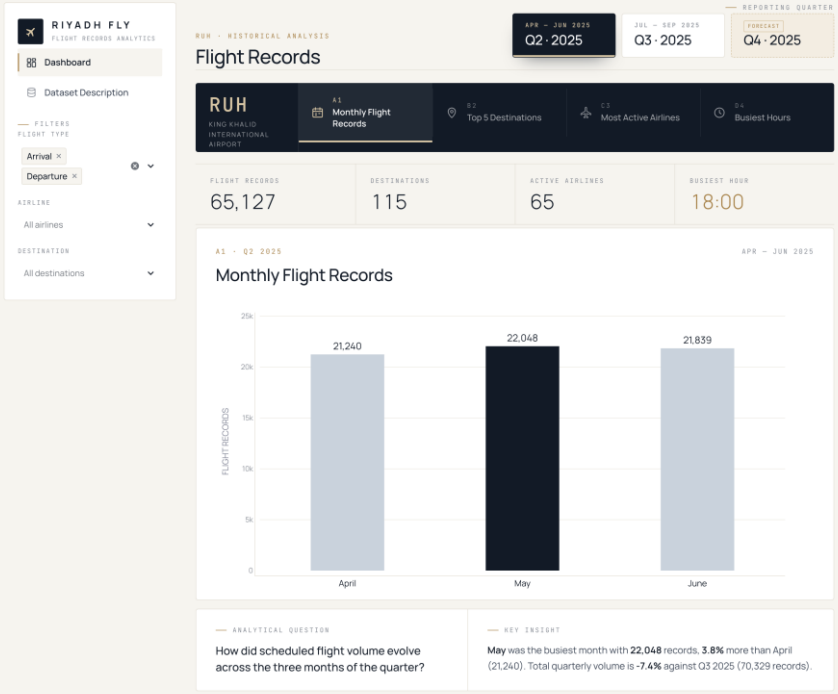


FIGURE 8.1

Q2 2025 Dashboard

The Q2 2025 dashboard summarizes flight activity at King Khalid International Airport, highlighting monthly flight records, destinations, active airlines, and the busiest hour. Users can select different views and apply interactive filters to explore flight data and identify key patterns.

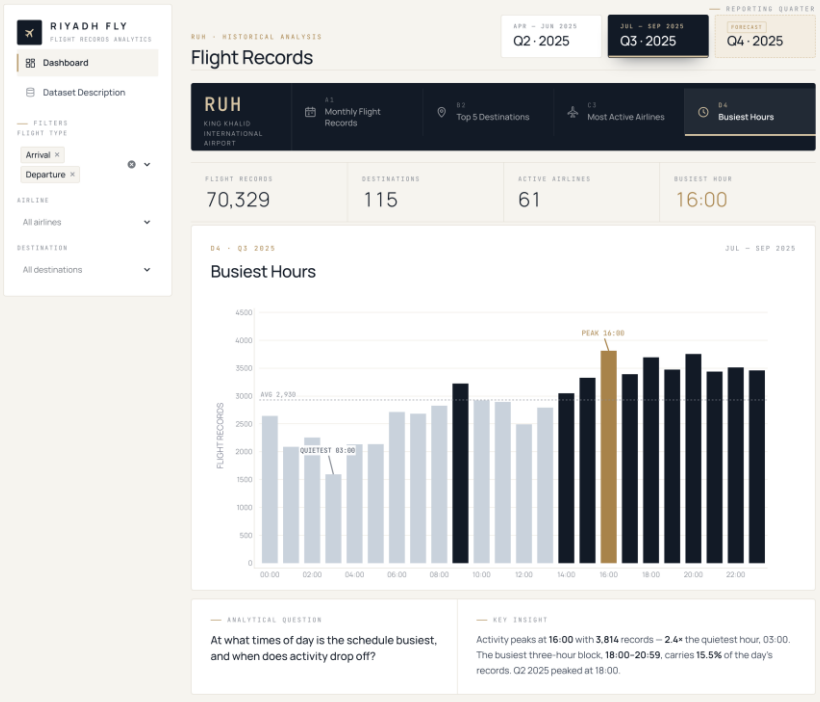


FIGURE 8.2

Q3 2025 Dashboard

The Q3 2025 dashboard summarizes flight activity at King Khalid International Airport, highlighting monthly flight records, destinations, active airlines, and the busiest hour. Users can select different views and apply interactive filters to explore flight data and identify key patterns.

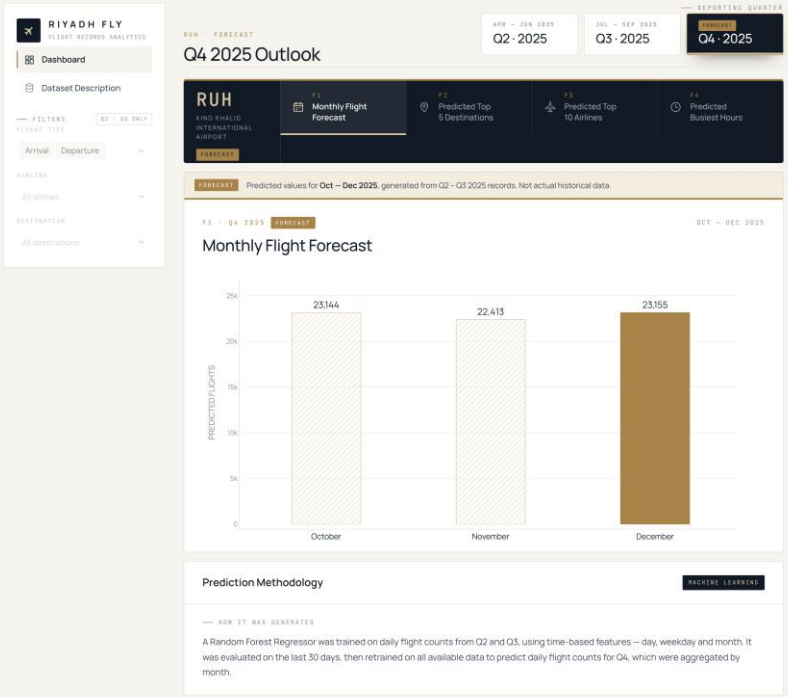


FIGURE 8.3

Q4 2025 Forecast dashboard

The Q4 2025 dashboard presents predicted flight activity, highlighting monthly flight forecasts, the top destinations and airlines, and the busiest hours. Users can explore different forecast views and understand the prediction methodology.

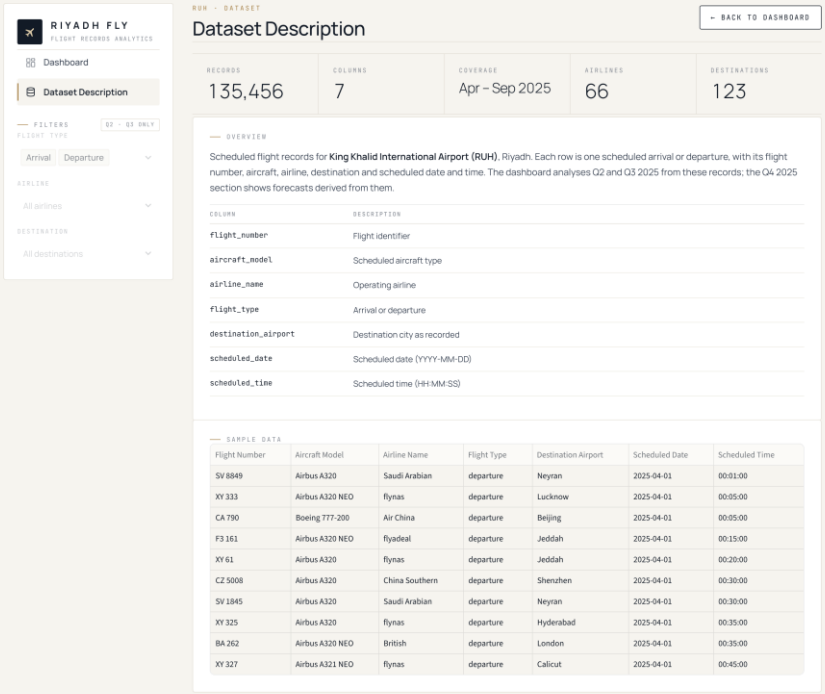


FIGURE 8.4

Dataset Description

The Dataset Description page provides an overview of the cleaned dataset, highlighting key statistics, column definitions, and sample flight records. It helps users understand the dataset structure and the information available for analysis.

— SECTION 09

Key Insights

The analysis of flight records from Q2 and Q3 2025 revealed several key patterns in flight activity at King Khalid International Airport.

01

Most Popular Destination

Jeddah consistently ranked as the most frequently served destination, including both departures from Riyadh to Jeddah and arrivals from Jeddah to Riyadh.

02

Daily Flight Activity

The dataset recorded an average of approximately 740 flight records per day across the analyzed period.

03

Busiest Hours

Flight activity was generally higher during the afternoon and evening, with peak hours varying between quarters.

04

Most Active Airline

Saudi Arabian Airlines (Saudia) consistently ranked as the airline with the highest number of flight records, leading other airlines by a substantial margin.

— SECTION 10

What I Learned

Through the Riyadh Fly project, I learned how to clean and analyze real-world flight data and gained a better understanding of aviation activity, including daily flight volumes, average flight rates, popular destinations, airline activity, and peak travel hours. I also developed skills in exploratory data analysis, data visualization, and applying Random Forest Regression to forecast flight activity. Additionally, I learned how to build an interactive Streamlit dashboard to present findings clearly and support a better understanding of flight patterns and trends.

— SECTION 11

Limitations of the Dataset

The dataset has several limitations that may affect the analysis and forecasting results. First, Q1 data is unavailable, limiting the analysis to Q2 and Q3 of 2025 and preventing a complete annual comparison. Second, the dataset covers a relatively short period, which may not capture long-term trends or seasonal variations. Third, some information, such as terminal and time zone details, may be missing, limiting the analysis of certain operational aspects. Additionally, the dataset provides scheduled flight information, which may not fully reflect actual flight times or delays.

— SECTION 12

Recommendations for Future Analysis

Future analysis could be improved by collecting flight data over a longer period, including Q1 data and additional years, to identify long-term trends and seasonal patterns. Including actual departure and arrival times would help analyze flight delays and compare scheduled flights with actual operations. Additional information, such as flight duration, passenger numbers, aircraft capacity, and weather conditions, could provide deeper insights into airport activity. Furthermore, future work could evaluate different machine learning models and compare their performance to improve forecasting accuracy. Finally, the Streamlit dashboard could be enhanced with additional filters, more detailed visualizations, and updated datasets to support more comprehensive analysis.

— SECTION 13

Conclusion

The Riyadh Fly project applied data cleaning, exploratory data analysis, data visualization, and predictive modeling to examine flight activity at King Khalid International Airport during Q2 and Q3 of 2025. The analysis identified important patterns, including Jeddah as the most frequently served destination, Saudia as the most active airline, an average of approximately 740 flight records per day, and variations in flight activity throughout the day. A Random Forest Regressor was used to estimate daily flight counts and hourly activity for Q4 2025, while historical averages were used to estimate destination and airline activity. These results were presented through an interactive Streamlit dashboard, making the findings easier to explore and understand. Overall, the project provided practical experience in data analysis, visualization, forecasting, and dashboard development while demonstrating how raw flight records can be transformed into meaningful insights. The findings and forecasts should be interpreted in light of the dataset's limitations and the use of historical patterns for prediction.

— SECTION 14

References

01 Alsubaie, M. (n.d.). *King Khalid International Airport Flights Dataset*. Kaggle. https://www.kaggle.com/datasets/mohammedalsubaie/king-khalid-international-airport-flights-dataset/data